



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 8 • STANDARD 6.DS.6D

Appropriate Measures

Lesson 8-4 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____



TODAY'S OBJECTIVES

Content Objective: I can choose the best measure of center for a data set based on its shape.

Language Objective: I can explain my choice using the words mean, median, outlier, and skewed.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mean

Median

Outlier

Skewed

Data distribution

Variability



Tap any word to see what it means and a picture.

1 The average of a data set is the .

2 The middle value when data is in order, often best when there is an outlier, is the .

3 A value far from the rest of the data, which can pull the mean, is an .

4 Data that is bunched on one side with a tail on the other is .

5 The way data values are spread out is the .

6 How much the data values differ is the .

Reflect — Exit Ticket

Data set: 15, 18, 16, 17, 15, 72. Which measure of center best represents the data?

- A. Median, because 72 is an outlier
- B. Mean, because it uses all values
- C. Mean, because it is always best
- D. Neither works for this data

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Mean
2. **Notes 2:** Median
3. **Notes 3:** Outlier
4. **Notes 4:** Skewed
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Appropriate Measures is skipping the key idea: "Use the mean for data with no outliers; use the median when an outlier or a skewed shape would pull the mean away from typical." — always check your work against this rule before you submit.*
8. **Try It 1:** A. Mode, because it names the value that appears most — *The mode is the value that shows up most often. Here size M appears 4 times, so the mode (M) tells the manager what to order most.*
9. **Try It 2:** A. Range, because it is the highest minus the lowest value — $\text{Range} = \text{max} - \text{min} = 18 - 6 = 12$. *Because it uses only the two most extreme values, one outlier (18) changes the range a lot.*
10. **Exit Ticket:** A. Median, because 72 is an outlier — *The value 72 is an outlier. The mean (25.5) is pulled high by 72 and doesn't represent the typical values. The median (16.5) is a better measure of center.*